

Serious Adverse Events Associated with GLP-1 Receptor Agonists: A Pharmacovigilance Analysis of the FDA FAERS Database

Background

Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are a class of medications widely used for the treatment of type 2 diabetes mellitus and, more recently, obesity. These drugs improve glycemic control by stimulating insulin secretion, suppressing glucagon release, and delaying gastric emptying. In addition to their glucose-lowering effects, GLP-1 receptor agonists promote weight loss and have demonstrated cardiovascular benefits in patients with diabetes. Because of these therapeutic advantages, the clinical use of GLP-1 medications such as semaglutide, liraglutide, dulaglutide, and tirzepatide has increased significantly worldwide (Filippatos et al., 2014).

Despite their effectiveness, the expanding use of GLP-1 receptor agonists has raised concerns regarding their safety profile. Clinical trials have documented several common adverse effects associated with these medications, including nausea, vomiting, and gastrointestinal discomfort. Although many of these reactions are mild or transient, rare but serious complications have also been reported, emphasizing the need for continued monitoring of these medications after they enter the market (Filippatos et al., 2014). Post-marketing safety surveillance plays a critical role in identifying rare or delayed adverse events that may not be observed during pre-approval clinical trials.

As the use of these medications increases, monitoring their safety in real-world settings becomes increasingly important. One important source of post-marketing safety data is the U.S. Food and Drug Administration Adverse Event Reporting System (FAERS), which collects reports of

adverse drug reactions submitted by healthcare professionals, manufacturers, and patients.

Analyzing FAERS data allows researchers to identify patterns of adverse events and potential safety signals associated with medications after they reach the market.

Literature Review

Pharmacovigilance has become an essential component of modern drug safety monitoring.

Spontaneous reporting systems such as the U.S. Food and Drug Administration Adverse Event Reporting System (FAERS) allow researchers to evaluate adverse drug reactions in real-world populations. FAERS collects reports submitted by healthcare professionals, pharmaceutical manufacturers, and patients regarding suspected adverse drug reactions. Because the database includes large numbers of reports from diverse patient populations, it provides valuable insights into drug safety signals that may not be captured in controlled clinical trials.

Several recent studies have utilized FAERS to investigate the safety profile of GLP-1 receptor agonists. For example, Osei et al. (2024) examined gastrointestinal safety outcomes associated with GLP-1 medications and reported increased reporting frequencies for adverse events such as nausea, vomiting, and delayed gastric emptying. These findings are consistent with earlier clinical studies demonstrating that gastrointestinal symptoms are among the most frequently reported side effects of GLP-1 receptor agonists.

In addition to gastrointestinal effects, other pharmacovigilance studies have explored metabolic and nutritional adverse events associated with GLP-1 therapies. He et al. (2024) conducted a large FAERS-based pharmacovigilance analysis examining metabolic and nutritional disorders reported among patients receiving GLP-1 receptor agonists. Their findings identified several potential safety signals, including dehydration and metabolic disturbances, particularly among

individuals receiving semaglutide and liraglutide. These results suggest that differences in adverse event profiles may exist across specific GLP-1 medications.

Research has also examined the occurrence of more severe outcomes associated with GLP-1 receptor agonists. Bhattacharyya et al. (2024) analyzed FAERS data to evaluate mortality and serious adverse events associated with these medications. Their study identified several safety signals related to serious outcomes, highlighting the importance of continued post-marketing monitoring. Such findings underscore the need for further investigation into the characteristics and predictors of serious adverse events among individuals using GLP-1 drugs.

Other studies have investigated specific categories of adverse reactions associated with GLP-1 therapies. For example, Chen et al. (2025) evaluated neurological adverse events reported in FAERS for GLP-1 receptor agonists and detected several potential neurological safety signals that may warrant further clinical investigation. Similarly, research analyzing European pharmacovigilance databases has explored psychiatric adverse events linked to GLP-1 medications, suggesting that certain neurological or psychiatric outcomes may occur in rare cases.

Overall, existing research demonstrates that GLP-1 receptor agonists are generally well tolerated but may be associated with a range of adverse events, including gastrointestinal symptoms, metabolic disturbances, and rare serious complications. Although pharmacovigilance studies have provided important insights into the safety of these medications, additional analyses are needed to better understand the distribution and predictors of serious adverse events reported for GLP-1 receptor agonists. Given the rapid growth in the use of these medications for both diabetes and weight management, further investigation using large real-world datasets such as

FAERS is essential for improving the understanding of drug safety patterns.

Study Aim

To describe and compare serious adverse events reported for GLP-1 receptor agonists in the FAERS database and to explore whether report characteristics such as drug type, age, and sex are associated with serious outcomes.

Research Questions

RQ1: What types of serious outcomes (e.g., hospitalization, death, life-threatening events) are reported for GLP-1 receptor agonists in FAERS, and how frequently does each outcome occur?

RQ2: Which adverse events are most frequently associated with serious outcomes among GLP-1 drug reports?

RQ3: Do serious adverse event patterns differ across individual GLP-1 drugs?

Hypotheses

H1: Hospitalization will be the most frequently reported serious outcome among GLP-1 adverse event reports.

H2: The distribution of serious adverse events will differ across GLP-1 drugs.

Data Source

This study utilized data from the U.S. Food and Drug Administration Adverse Event Reporting System (FAERS), a publicly available pharmacovigilance database that collects reports of suspected adverse drug reactions submitted by healthcare professionals, manufacturers, and

patients. FAERS is widely used to identify potential drug safety signals and to monitor post-marketing drug safety.

Study Design

A retrospective pharmacovigilance analysis was conducted using FAERS reports associated with glucagon-like peptide-1 receptor agonists (GLP-1 RAs). Reports from the FAERS dataset were filtered to identify cases involving GLP-1 medications including semaglutide, liraglutide, dulaglutide, exenatide, lixisenatide, and tirzepatide products.

Data Processing

Data were cleaned and processed using the R statistical programming language. Duplicate reports were removed based on case identifiers, and variables including drug name, adverse reaction terms, outcome categories, and seriousness indicators were extracted. Descriptive statistics were calculated to summarize the distribution of adverse events and serious outcomes.

Statistical Analysis

Descriptive analyses were performed to examine the frequency and distribution of adverse event reports. Counts and percentages were calculated for adverse reactions, serious outcomes, and drug-specific reports. Visualizations were created using the ggplot2 package in R to illustrate reporting patterns across GLP-1 medications.

Results:

Table 1. Top Adverse Reactions Reported

PT	N	Percent
Incorrect dose administered	12486	12.88
Nausea	11960	12.34
Vomiting	7906	8.16
Diarrhoea	7384	7.62
Off label use	5904	6.09
Injection site pain	5629	5.81
Constipation	5551	5.73
Impaired gastric emptying	5177	5.34
Extra dose administered	4320	4.46
Decreased appetite	3621	3.74
Drug ineffective	3354	3.46
Abdominal pain	3288	3.39
Abdominal pain upper	3084	3.18
Dehydration	2696	2.78
Fatigue	2660	2.74
Weight increased	2535	2.62
Product dose omission issue	2497	2.58
Accidental underdose	2366	2.44
Dyspepsia	2320	2.39
Headache	2189	2.26

Table 1 presents the most frequently reported adverse reactions associated with GLP-1 receptor agonists in the FAERS 2025 dataset. The most commonly reported reactions included **incorrect dose administered, nausea, vomiting, and diarrhoea**. Gastrointestinal symptoms such as nausea, vomiting, and diarrhoea are consistent with the known pharmacologic effects of GLP-1 receptor agonists, which slow gastric emptying and affect appetite regulation.

Reports related to **incorrect dose administration and off-label use** likely reflect prescribing practices, medication errors, or real-world medication use patterns rather than direct

pharmacologic toxicity. Overall, these findings highlight the predominance of gastrointestinal adverse events among GLP-1 therapies in spontaneous reporting systems.

Table 2. Distribution of Serious Outcomes

Outcome	N	Percent
Other serious outcome	72249	58.24
Hospitalization	40258	32.45
Life-threatening	4122	3.32
Disability	4025	3.24
Death	2171	1.75
Required intervention	1124	0.91
Congenital anomaly	113	0.09

Table 2 summarizes the distribution of serious outcomes reported in FAERS for GLP-1 drugs.

The most frequently reported serious outcome was **other medically significant outcomes**, followed by **hospitalization**. Less frequently reported outcomes included **life-threatening events, disability, and death**.

These findings suggest that while many adverse event reports involve clinically significant outcomes requiring hospitalization or medical intervention, the proportion of fatal outcomes remains relatively small compared with other categories. The distribution of outcomes highlights the range of severity captured within pharmacovigilance reporting systems.

Table 3. Distribution of FAERS Reports by GLP-1 Drug

DRUGNAME	N	Percent
MOUNJARO	71121	28.4
OZEMPIC	62870	25.1
ZEPBOUND	56343	22.5
WEGOVY	18996	7.58
TRULICITY	16749	6.69
SEMAGLUTIDE	5302	2.12
RYBELSUS	4684	1.87
TIRZEPATIDE	3396	1.36
BYDUREON BCISE	3061	1.22
VICTOZA	2516	1

SAXENDA	2151	0.86
SOLIQUA 100/33	781	0.31
XULTOPHY 100/3.6	618	0.25
LIRAGLUTIDE	534	0.21
BYETTA	486	0.19
DULAGLUTIDE	138	0.06
INSULIN GLARGINE\LIXISENATIDE	111	0.04
EXENATIDE	106	0.04
CYANOCOBALAMIN\SEMAGLUTIDE	88	0.04
BYDUREON	84	0.03
METHYLCOBALAMIN\SEMAGLUTIDE	44	0.02
CYANOCOBALAMIN\TIRZEPATIDE	38	0.02
GLYCINE\TIRZEPATIDE	34	0.01
METHYLCOBALAMIN\NADIDE\TIRZEPATIDE	27	0.01
PYRIDOXINE\TIRZEPATIDE	25	0.01
CYANOCOBALAMIN\GLYCINE\TIRZEPATIDE	25	0.01
NIACINAMIDE\TIRZEPATIDE	22	0.01
LIXISENATIDE	21	0.01
LEVOCARNITINE\TIRZEPATIDE	18	0.01
PYRIDOXINE\SEMAGLUTIDE	14	0.01
CYANOCOBALAMIN\GLYCINE\SEMAGLUTIDE	12	0
GLYCINE\SEMAGLUTIDE	10	0
CYANOCOBALAMIN\LEVOCARNITINE\SEMAGLUTIDE	7	0
LEVOCARNITINE\SEMAGLUTIDE	4	0
METHYLCOBALAMIN\TIRZEPATIDE	3	0
GLYCINE\METHYLCOBALAMIN\TIRZEPATIDE	3	0
GLYCINE\METHYLCOBALAMIN\SEMAGLUTIDE	3	0
INSULIN DEGLUDEC\LIRAGLUTIDE	1	0
CHOLINE\INOSITOL\METHIONINE\METHYLCOBALAMIN\TIRZEPATIDE	1	0
CHROMIUM		
PICOLINATE\CYANOCOBALAMIN\LEVOCARNITINE\SEMAGLUTIDE	1	0
CYANOCOBALAMIN\NADIDE\SEMAGLUTIDE	1	0

Table 3 displays the number of adverse event reports associated with individual GLP-1 receptor agonist medications. The largest number of reports were associated with **Mounjaro, Ozempic, and Zepbound**, followed by **Wegovy and Trulicity**.

This pattern likely reflects the **rapid increase in prescribing and widespread clinical use of newer GLP-1 drugs**, particularly semaglutide and tirzepatide products, during recent years.

Increased public awareness, expanding indications for weight management, and media attention may also contribute to higher reporting frequencies for these medications in FAERS.

Table 4. Serious vs Non-Serious Report Counts

SERIOUS_FLAG	N	Percent
0	126387	50.46
1	124062	49.54

Table 4 compares the number of **serious and non-serious adverse event reports** in the dataset. The results indicate that approximately **half of the reports were classified as serious events**, including hospitalization, disability, life-threatening conditions, or death.

The relatively high proportion of serious reports reflects the nature of FAERS as a pharmacovigilance system where healthcare professionals and patients are more likely to report events perceived as clinically significant. However, because FAERS is a **spontaneous reporting system**, these counts do not represent true incidence rates and should be interpreted as reporting patterns rather than measures of risk.

Overall, the analysis identified gastrointestinal symptoms as the most frequently reported adverse reactions associated with GLP-1 receptor agonists. The most common reactions included nausea, vomiting, diarrhea, and impaired gastric emptying. Hospitalization and other medically significant outcomes represented the most frequently reported serious outcomes in the FAERS dataset. Additionally, adverse event reports were heavily concentrated among newer GLP-1 medications such as semaglutide and tirzepatide products. These findings demonstrate how pharmacovigilance databases can provide valuable insights into real-world drug safety patterns and help identify potential adverse event signals associated with widely used medications.

Figure 1. Distribution of Serious Adverse Outcomes Reported for GLP-1 Drugs (FAERS 2025)

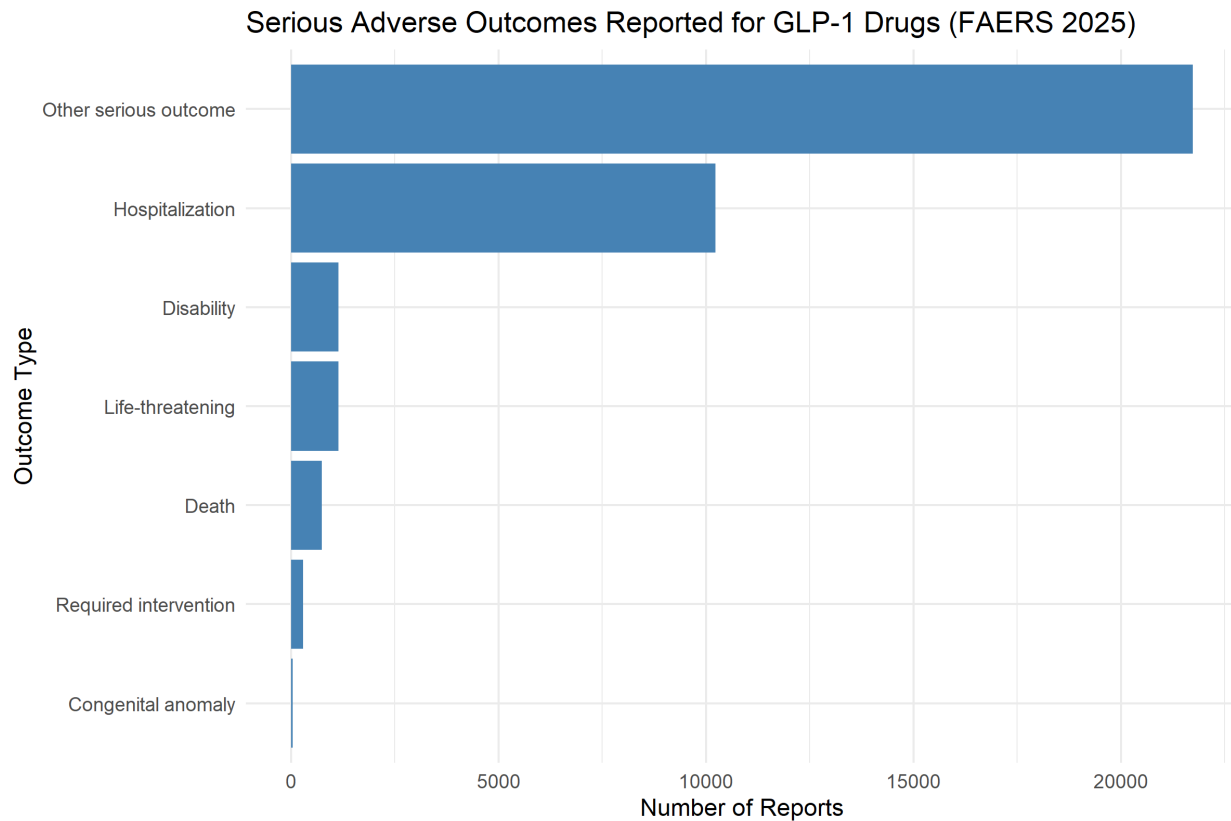


Figure 1 presents the distribution of serious adverse outcomes associated with GLP-1 drugs reported in the FAERS database in 2025. The majority of reports were categorized as **“other serious outcomes,”** followed by **hospitalization**, indicating that many adverse events were clinically significant and required medical attention. Reports of **life-threatening events, disability, and death** were comparatively less frequent, while congenital anomalies were rarely reported.

Figure 2. Distribution of Serious Outcomes by GLP-1 Drug

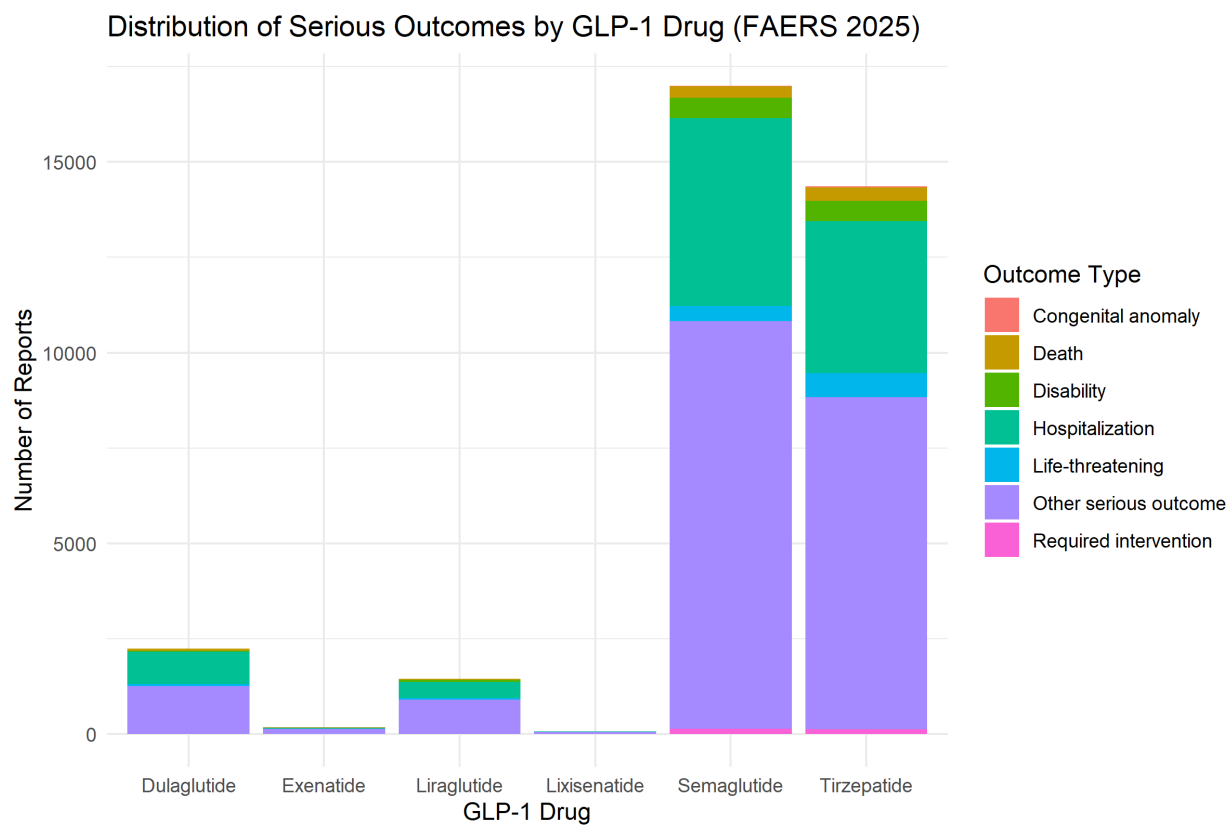


Figure 2 illustrates the distribution of serious adverse outcomes across different GLP-1 drugs.

Semaglutide and tirzepatide accounted for the largest number of serious adverse event reports, reflecting their widespread use and increasing prescription rates. Across all drugs, the most commonly reported outcomes were **hospitalization and other serious medical events**, while life-threatening events and deaths represented a smaller proportion of reports.

Figure 3. Top 25 Adverse Reactions Among Serious GLP-1 Reports

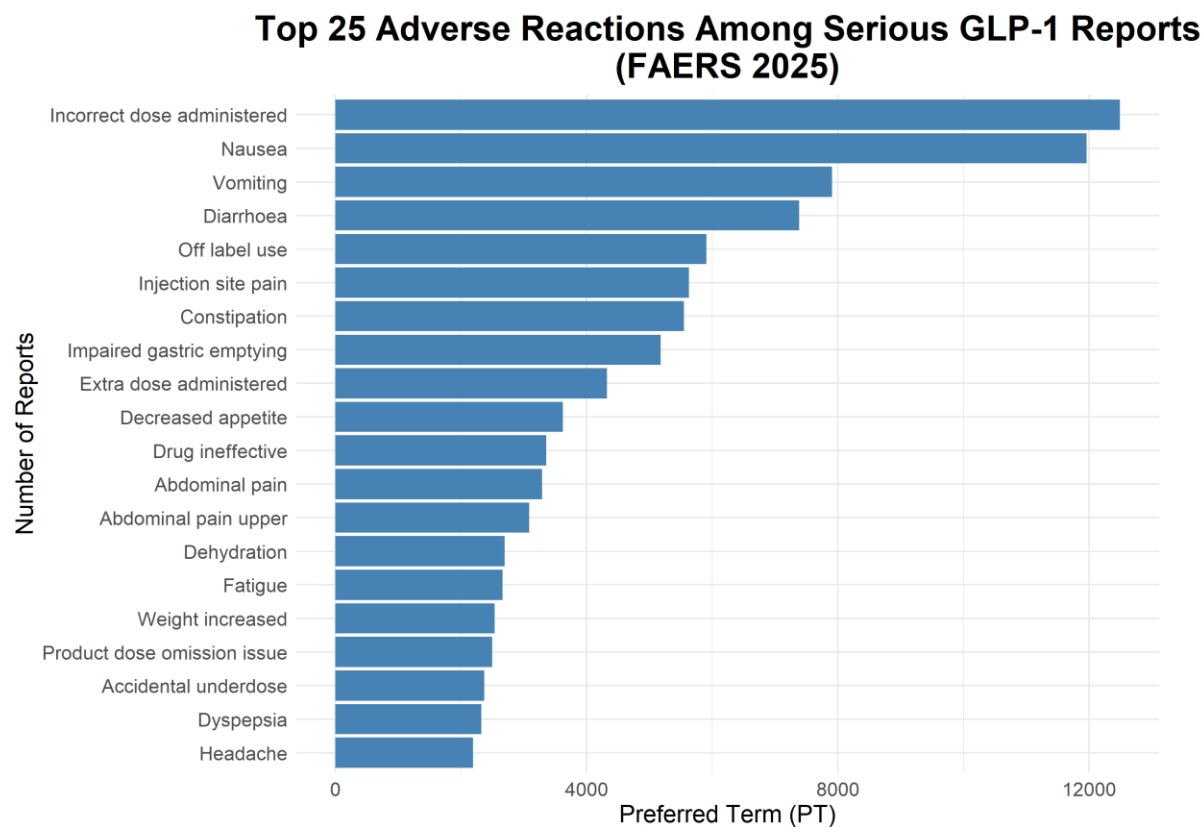


Figure 3 displays the most frequently reported adverse reactions associated with serious GLP-1 reports in the FAERS database. The most common reactions included **incorrect dose administered, nausea, vomiting, and diarrhea**, followed by **off-label use and injection site pain**. Several gastrointestinal symptoms such as **impaired gastric emptying, abdominal pain, and dyspepsia** were also frequently reported, highlighting the predominance of gastrointestinal-related adverse events among GLP-1 drug users.

Figure 4. Most Reported GLP-1 Drugs in FAERS 2025

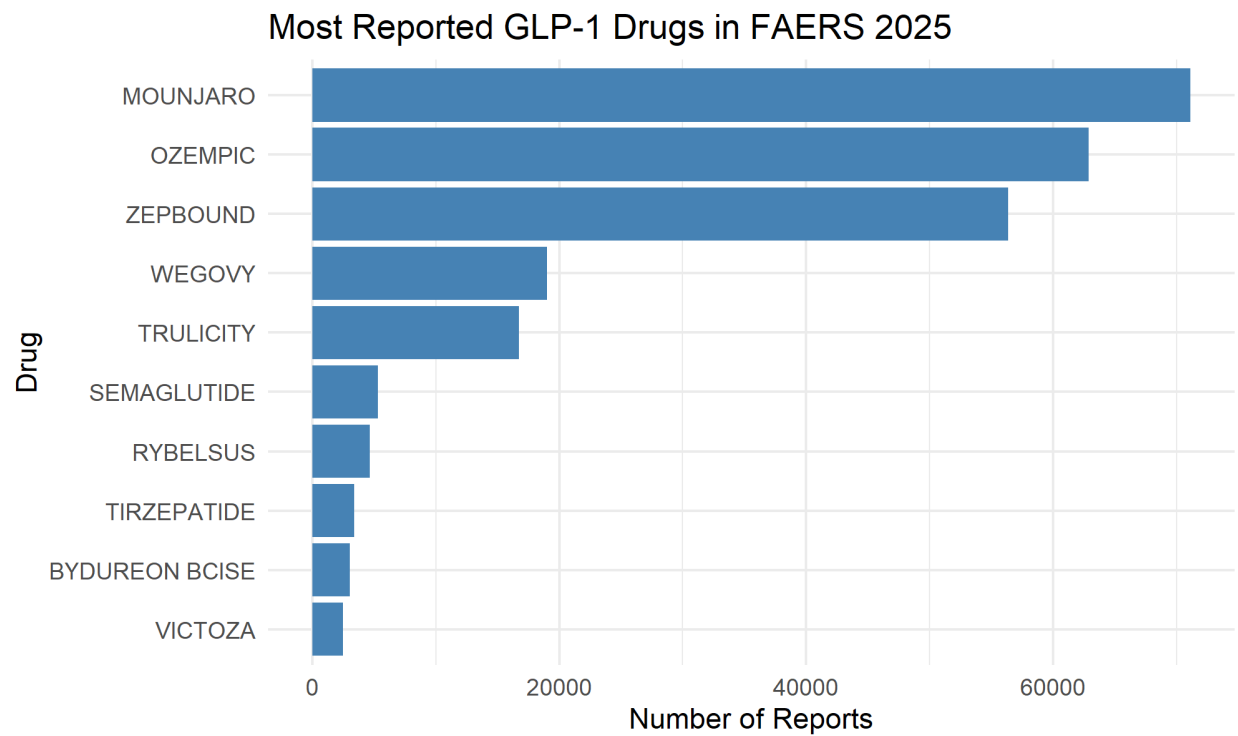


Figure 4 shows the distribution of reported adverse events by GLP-1 drug. **Mounjaro (tirzepatide), Ozempic (semaglutide), and Zepbound were the most frequently reported drugs** in the FAERS dataset, accounting for the largest proportion of adverse event reports. Older GLP-1 medications such as **Victoza and Bydureon** contributed relatively fewer reports, which may reflect differences in prescribing patterns and market adoption.

Figure 5. Serious Outcomes Associated with GLP-1 Adverse Events

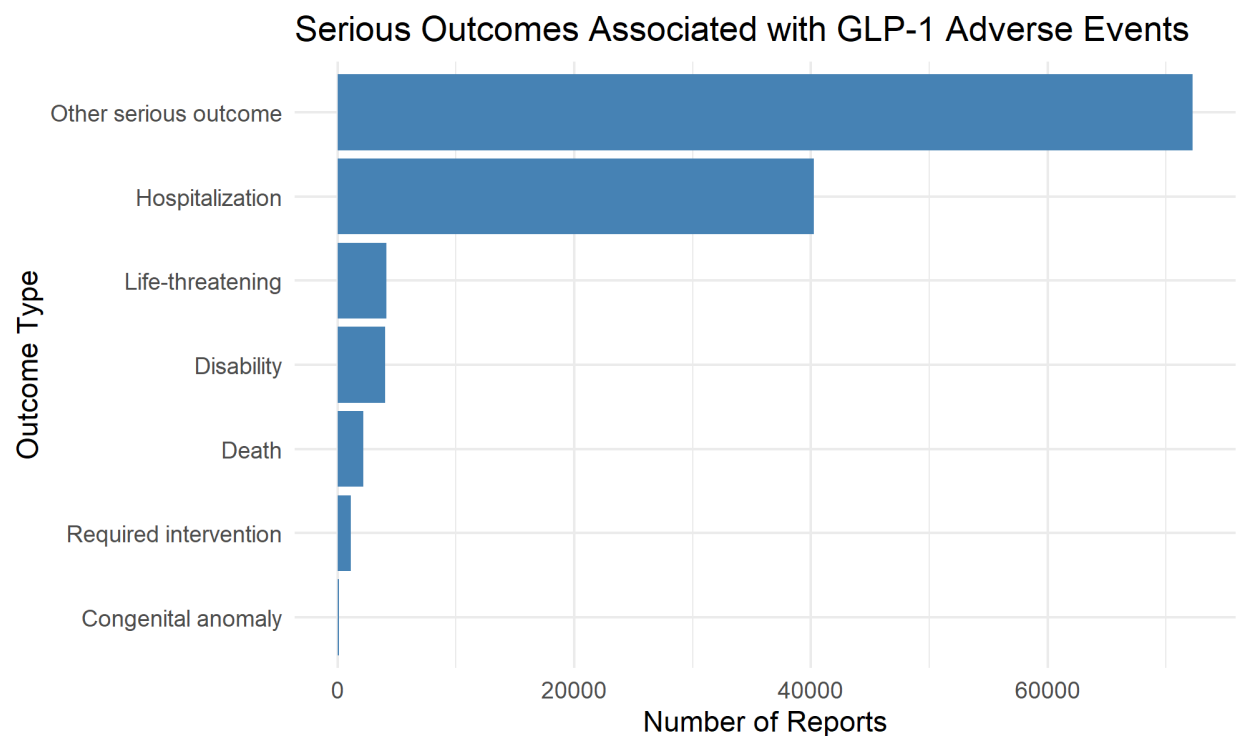


Figure 5 summarizes the overall distribution of serious outcomes associated with GLP-1 adverse events. The largest category was **other serious medical outcomes**, followed by **hospitalization**, suggesting that many reported events required clinical evaluation or intervention. In contrast, **life-threatening events, disability, and death were relatively uncommon**, indicating that while adverse reactions occur, the most severe outcomes are less frequently reported.

Figure 6. Top 20 Adverse Events Associated with GLP-1 Drugs

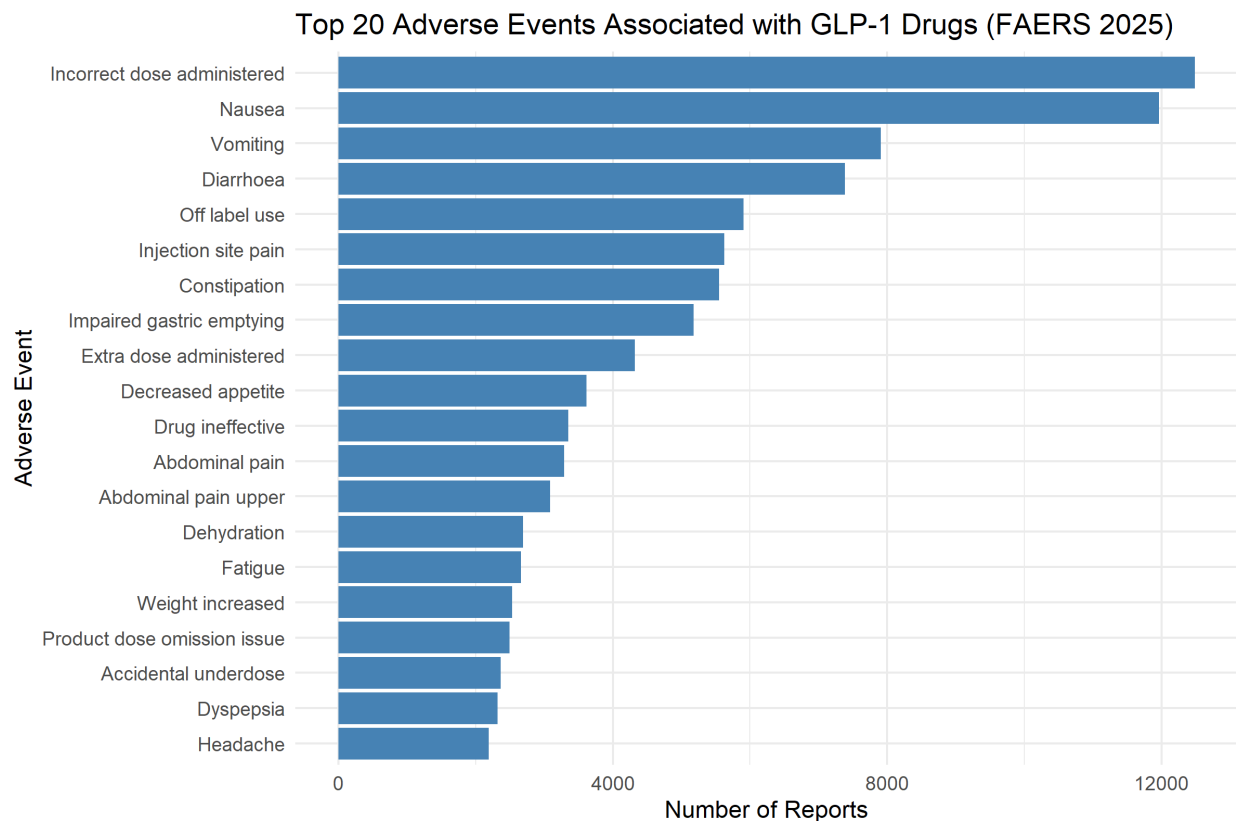


Figure 6 presents the most commonly reported adverse events associated with GLP-1 drugs in FAERS. The leading events included **incorrect dose administered, nausea, vomiting, and diarrhea**, followed by **off-label use and injection site pain**. Additional frequently reported events included **constipation, impaired gastric emptying, and decreased appetite**, which are consistent with the known gastrointestinal side-effect profile of GLP-1 receptor agonists.

Figure 7. Proportion of Serious Outcomes by GLP-1 Drug

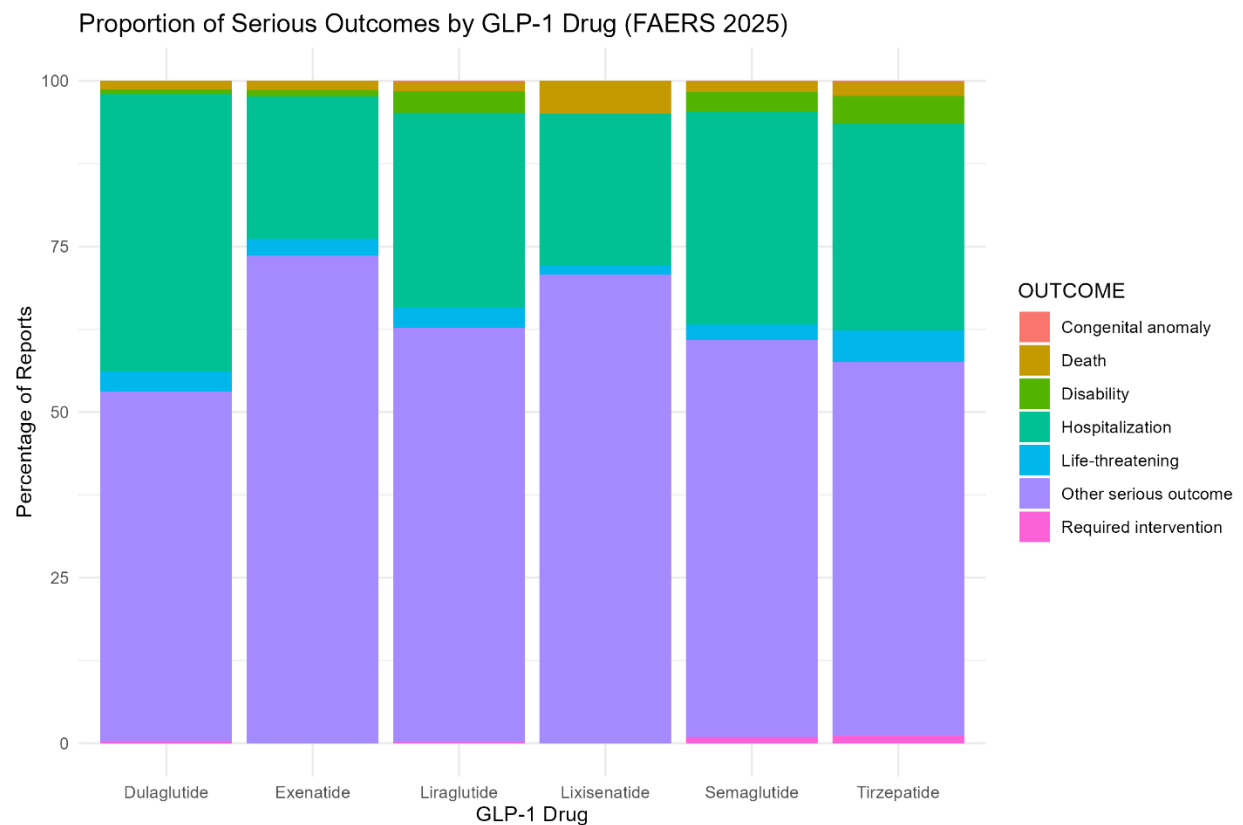


Figure 7 illustrates the proportion of different serious outcomes across GLP-1 drugs. Although the absolute number of reports varied between medications, the **distribution of outcome types remained relatively consistent across drugs**. The majority of reports were categorized as **other serious outcomes or hospitalization**, while **death and life-threatening events represented a small proportion of total reports**.

Overall, adverse event reports were heavily concentrated among semaglutide and tirzepatide products, reflecting their widespread clinical use and increased public attention in recent years.

Discussion

The findings of this pharmacovigilance analysis provide insight into real-world adverse event reporting patterns associated with GLP-1 receptor agonists. Gastrointestinal symptoms such as nausea, vomiting, and diarrhea were among the most frequently reported adverse reactions. These results are consistent with previous pharmacovigilance and clinical studies that have identified gastrointestinal disturbances as the most common side effects associated with GLP-1 therapies (Osei et al., 2024; Filippatos et al., 2014).

The analysis also demonstrated that most serious adverse event reports were categorized as other medically significant outcomes or hospitalization. Although life-threatening events and death were reported, they represented a relatively small proportion of the overall reports. These findings align with previous FAERS analyses that have suggested that while GLP-1 receptor agonists are generally well tolerated, continued monitoring is important to identify rare but serious adverse events (Bhattacharyya et al., 2024).

Additionally, adverse event reports were heavily concentrated among semaglutide and tirzepatide products. This pattern likely reflects the rapid expansion in the use of these medications for both diabetes management and weight loss, as well as increased public awareness and prescribing patterns.

Limitations

This analysis relies on data from the FAERS spontaneous reporting system, which is subject to several important limitations. Reports may be affected by underreporting, stimulated reporting,

and reporting bias. Additionally, FAERS data do not contain complete clinical information and cannot establish causal relationships between a drug and an adverse event. The number of reports should therefore be interpreted as reporting patterns rather than true incidence rates. Despite these limitations, FAERS remains a valuable tool for identifying potential safety signals and generating hypotheses for further clinical investigation.

Conclusion

This pharmacovigilance analysis of FAERS data provides an overview of adverse event reporting patterns associated with GLP-1 receptor agonists. The results indicate that gastrointestinal symptoms represent the most frequently reported adverse reactions, while hospitalization and other serious medical outcomes are the most common serious outcomes reported. Adverse event reports were concentrated among newer GLP-1 medications such as semaglutide and tirzepatide, likely reflecting their widespread clinical use. These findings are consistent with previous pharmacovigilance studies reporting gastrointestinal symptoms as the most common adverse reactions associated with GLP-1 receptor agonists (Osei et al., 2024; Filippatos et al., 2014). Continued pharmacovigilance monitoring is important as the use of GLP-1 therapies continues to expand globally.

References

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